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Structural evolution from sandwich to cycle enhances CO_2 Cycloaddition catalysis in Zn^{II}-Tb^{III} complexes

Xue-Qin Song *, Peng-Li Zhao, Yi-Yi Yang, Peng-Fei Qiu

School of Chemistry and Chemical Engineering, Lanzhou Jiaotong University, Lanzhou, 730070, China

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A B S T R A C T

To advance fundamental understanding of structure-activity relationships of catalysts for CO_2 cycloaddition, we developed two novel Zn^{II}-Tb^{III} heterometallic complexes, based on a multidentate asymmetric ligand (**H_2L**), where salicylamide and 4-nitrosalicylaldimide are spaced by 1,2-bis(2-ethoxy)ethyl. $[Zn_2Tb_2L_2(\mu_3\text{-}OH)_2(NO_3)_4]\cdot 6CH_3CN$ (**ZnTb-1**) and $[Zn_3Tb_2L_4(NO_3)_4(H_2O)_2]\cdot 2CH_3CN\cdot H_2O$ (**ZnTb-2**) were synthesized in a solvent-directed manner. In pure CH_3CN, the heterometallic sandwich structure (**ZnTb-1**) is formed, wherein two **L^{2-}** with coordination mode of μ_3-κ^2O1:κO2:κN2:κO5 supported by an in-situ-generated μ_3-OH group bridge two Tb^{III} and two Zn^{II} ions. Upon switching to CH_3OH-CH_3CN mixture, four **L^{2-}** utilize a combination of μ_3-κO1:κO2:κO4:κN2:κO5 and μ_3-κ^2O1:κO2:κN2:κO5 coordination modes to anchor two Tb^{III} and three Zn^{II} ions, resulting in the formation of an interlocking cyclic compound **ZnTb-2**. The structural variation leads to dramatic catalytic performance with the observation of **ZnTb-2** achieving much higher CO_2 cycloaddition efficiency under identical mild conditions. Detailed analysis reveals that **ZnTb-2**, which integrates confined cavity, more exposed metal centers and Lewis base sites ensures sufficient interaction space between CO_2 and substrates. This work demonstrates how precise heterometallic complex engineering enables near-quantiative CO_2 conversion, shedding some light on the development of sustainable Zn^{II}-Tb^{III} heterometallic catalytic systems.

1. Introduction

Combating climate change and establishing a sustainable ecosystem are among the most critical challenges of the 21st century. In this endeavor, carbon capture and utilization (CCU) have emerged as a pivotal strategy, especially as global initiatives accelerating toward carbon neutrality [1]. Given that CO_2 is the primary greenhouse gas driving global warming, its transformation into value-added chemicals not only reduce atmospheric CO_2 levels but also leverage it as an abundant, inexpensive, and environmentally benign carbon feedstock. Among various CCU methods, the catalytic cycloaddition of CO_2 to epoxides stands out due to its 100 % atom economy and the broad industrial application of cyclic carbonates in lithium-ion batteries, pharmaceuticals and fine chemicals [2,3]. However, the inherent thermodynamic stability and kinetic inertness of CO_2 necessitate highly efficient catalysts to drive such transformations under mild conditions [4]. The catalytic performance in CO_2 cycloaddition hinges on three critical steps: epoxide activation, ring opening and CO_2 insertion. Thus, developing catalysts that synergistically optimize these steps is paramount. While significant progress has been made on diverse catalytic systems, including coordination polymers (CPs) [5] metal-organic frameworks (MOFs), [6–8] ionic liquids [9–11] covalent-organic frameworks (COFs) [12,13] and silica nanoparticles [14,15] these materials often face challenges in precisely controlling the spatial arrangement of multiple active sites at the molecular level. Given this, discrete heterometallic complex offer a unique platform for fundamental study, as their well-defined structures allow for precise investigation of multimetallic synergy. The integration of metals with distinct electronic properties can create multisite interactions for enhanced substrate activation, which are often unattainable in monometallic systems [16,17]. Furthermore, the use of non-precious metals makes such systems highly promising in terms of cost and sustainability. Previous studies have hinted that structural features in Zn^{II}-Ln^{III} assemblies, such as Lewis acid sites and adjacent Lewis base groups (e.g., –CONH–), play critical roles in catalysis by facilitating epoxide ring-opening and CO_2 insertion [18,19]. However, a clear mechanistic understanding of how the molecular structure of a discrete Zn^{II}-Ln^{III} complex dictates its catalytic performance is still lacking. Elucidating these structure-property

* Corresponding author.
E-mail address: songxueqin001@163.com (X.-Q. Song).

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relationships is therefore crucial for more efficient and sustainable CO_2 conversion catalysts.

In this context, two novel Zn^{II}-Ln^{III} heterometallic compounds were prepared based on a salicylamide imine multidentate ligand (**H_2L**) [20] to probe how structural features govern catalytic activity in CO_2 cycloaddition (Scheme 1). The two compounds have been well characterized by single-crystal X-ray diffraction, powder X-ray diffractometry (PXRD), infrared spectroscopy (IR), thermogravimetric analysis (TGA) and elemental analyses. Single crystal structural analysis reveals that two deprotonated **L^{2-}** in μ_3-**L**-κ^2O1:κO2:κN2:κ^2O5 mode are interlocked by two Tb^{3+} and two Zn^{2+} with the assistance of an in situ formed μ_3-OH to render **ZnTb-1** the sandwich-like structure. In contrast, four deprotonated **L^{2-}** in both μ_3-**L**-κO1:κO2:κO4:κN2:κO5 and μ_3-L-κ^2O1:κO2:κN2:κO5 mode are anchored by two Tb^{3+} and three Zn^{2+}, rendering **ZnTb-2** an interlocking cyclic structural feature. Remarkably, **ZnTb-2** achieves excellent efficiency in CO_2 cycloaddition under mild conditions, underscoring the profound impact of molecular architecture on catalytic performance. Mechanistic investigations demonstrate its superior activity arises from three key factors: more exposed metal centers enhances the epoxide activation and sequent ring opening, increased Lewis base sites improve CO_2 activation, and confined cavities promotes optimal substrate orientation and transition-state stabilization.

2. Materials and methods

The details of materials and instruments, X-ray crystallographic determination, general procedure for catalytic coupling of epoxides and CO_2, and computational details for catalytic mechanism were described in supporting information.

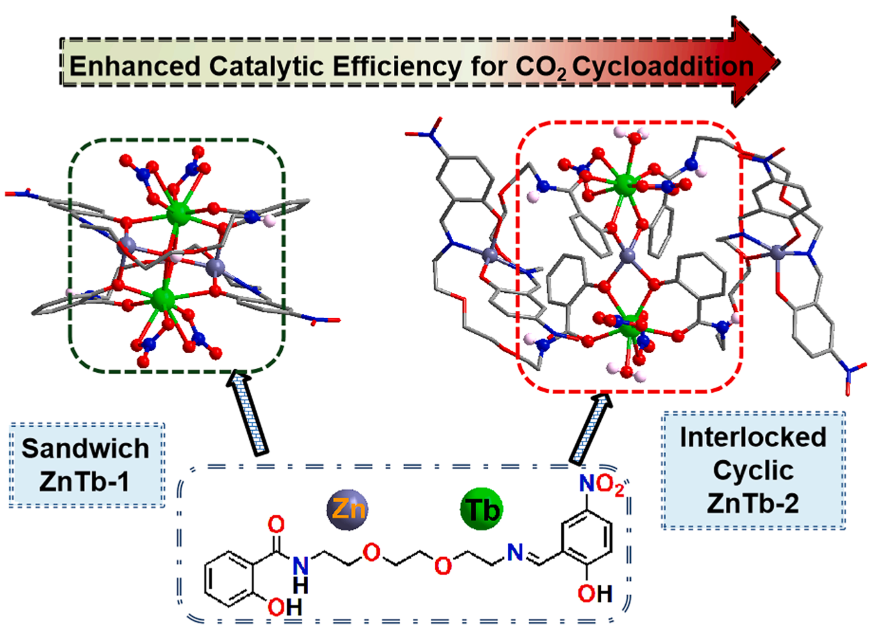
H_2L was prepared according to reference [20]. **ZnTb-1** was synthesised as following: 0.1 mmol **H_2L** (0.0417 g) and 0.2 mmol triethylamine (27 uL) were dissolved in 35 mL acetonitrile to obtain a bright yellow solution. After stirring for 30 min, 0.1 mmol (0.0297 g) Zn $(NO_3)_2{\cdot}6H_2O$ was added, and stirred for another 4 h to obtain a pale yellow suspension. To this suspension, 0.1 mmol $Tb(NO_3)_3{\cdot}6H_2O$ (0.0425 g) was added to obtain a clear solution which was stirred overnight. The resulted mixture was filtered into a sealed 10 – 20 mL glass vial and set aside for slow evaporation. After about two weeks, colorless block crystals suitable for single crystal X-ray diffraction were obtained. The crystalline samples were collected by filtration, washed with cold methanol, and dried in air. **ZnTb-2** was obtained by addition of 5 mL methanol to afford a clear solution before the addition of Tb $(NO_3)_3{\cdot}6H_2O$. Bulky samples for spectral characterization and sequently catalytic exploration were collected by repeated experiments.

$[Zn_2Tb_2L_2(\mu_3\text{-}OH)_2(NO_3)_4]{\cdot}6CH_3CN$ (**ZnTb-1**) Yield: 41.58 mg, 46 % based on $Tb(NO_3)_3{\cdot}6H_2O$. Elemental analysis for $C_{52}H_{62}N_{16}O_{28}Tb_2Zn_2$ (%): *Calcd*: C, 34.55; H, 3.46; N, 12.40.; *Found*: C, 34.23; H, 3.45; N, 14.43. IR (KBr, υ, cm^{-1}): 3450 (w), 1612 (s), 1558 (s), 1480 (s), 1312 (m), 1128 (m), 1021 (w), 821(w), 503 (w), 444(w).

$[Zn_3Tb_2L_4(NO_3)_4(H_2O)_2]{\cdot}2CH_3CN{\cdot}H_2O$ (**ZnTb-2**) Yield: 27.46 mg, 43 % based on $Tb(NO_3)_3{\cdot}6H_2O$. Elemental analysis for $C_{84}H_{96}N_{18}O_{43}Tb_2Zn_3$ (%): *Calcd*: C, 39.41; H, 3.78; N, 9.85.; *Found*: C, 39.29; H, 3.75; N, 9.81. IR (KBr, υ, cm^{-1}): 3452 (w), 3385 (w), 1614(s), 1558 (w), 1485 (w), 1297 (w), 1254 (m), 1103 (m), 1025 (s), 817 (w), 499 (w), 439 (w).

3. Results and discussion

Structure Description The reaction of fully deprotonated **H_2L** with 1 equiv. of $Zn(NO_3)_2{\cdot}6H_2O$ followed by the addition of 1 equiv. of Tb $(NO_3)_3{\cdot}6H_2O$ in acetonitrile and acetonitrile-methanol mixed solvent afford two different Zn^{II}-Ln^{III} heterometallic compounds, **ZnTb-1** and **ZnTb-2**. Each bulky crystalline samples were checked for crystalline purity by comparing the PXRD patterns with the calculated ones issued from the crystal data. As shown in Fig. S1, the experimental PXRD patterns of **ZnTb-1** and **ZnTb-2** are consistent with those simulated from their CIF files respectively, indicating their good phase purity. The crystal data information of **ZnTb-1** which is crystallized in triclinic P-1 space group as block crystals is shown in Table 1 and Table S1. There is one-half molecule constituted by one fully ligand **L^{2-}**, one Tb^{3+}, one Zn^{2+}, one in-situ formed μ_3-OH, two bidentate NO_3^- and three crystalline acetonitrile in the asymmetric unit. Driven by the coordination preference of metal ions, **L^{2-}** in **ZnTb-1** is also in open-loop configuration as in **Tb-2 [*c*]** based on **L^{2-}** [20]. However, two phenoxy oxygen, one amide oxygen, one imine nitrogen and one etheric oxygen of one **L^{2-}** in this case clamps one Tb^{3+} and two Zn^{2+} to afford a μ_3-**L**-κ^2O1:κO2:κO4:κN2:κ^2O5 new coordination mode (Mode I), where 1,2-bis(2-ethoxy)ethyl spacer is less compressed to afford the $N_{amide}{\bullet\bullet\bullet}N_{imine}$ distance of 7.462 Å, the distance of two benzene centers of 9.319 Å and the dihedral angle of 32.535° (Fig. 1a). The coordination sphere for Tb^{3+} are fulfilled by the salicylamide, phenoxy oxygen of 4-nitryl-salicylaldimine, one etheric oxygen of one **L^{2-}**, two μ_3-OH and two adjacent bidentate NO_3^- to result a



Scheme 1. Schematic illustration of **ZnTb-1** and **ZnTb-2** preparation as catalysts for CO_2 addition.

Table 1
Crystal data and structure refinement parameters for **ZnTb-1** and **ZnTb-2**.

	ZnTb-1	**ZnTb-2**
CCDC No	2484,801	2171,977
Empirical formula	$C_{52}H_{62}N_{16}O_{28}Tb_2Zn_2$	$C_{160}H_{176}N_{32}O_{84}Tb_4Zn_6$
Temperature (K)	293(2)	173.00(10)
formula_weight	4951.58	4919.22
Diffraction	MoKα (λ = 0.71073)	CuKα (λ = 1.54178)
Crystal system, Space group	Triclinic, P−1	Monoclinic, $P2_1/c$
Unit cell dimensions	a = 11.3631(6) Å	a = 33.5040(3) Å
	b = 12.7764(6) Å	b = 20.8029(2) Å,
	c = 13.1610(9) Å	c = 37.5290(4) Å
	α = 69.354(5)°	α = 90 °
	β = 75.202(5)°	β = 102.1800(10)°
	γ = 79.381(4)°	γ = 90°
V/Å^3, Z	1719.29(18), 1	25,568.2(4), 4
D_{calcd}/mg·m^{-3}, μ/mm^{-1}	1.746, 2.814	1.278, 6.604
F(000)	900	9888.0
θ range for data collection	3.293 °~25.999°	6.87 ° ~141.676°
index ranges, hkl	$-14 \le h \le 12$	$-41 \le h \le 40$
	$-15 \le k \le 15$	$-25 \le k \le 24$
	$-16 \le l \le 14$	$-43 \le l \le 45$
	$-15 \le k \le 15$	$-14 \le k \le 12$
	$-16 \le l \le 14$	$-14 \le l \le 12$
Independent reflections (R_{int})	0.0494	0.0518
Completeness	99.7 %	98.1 %
Reflections unique / collected	6731/11,032	48,333/109,092
Data / restraints / params	6731/0/458	48,333/153/2633
Goodness-of-fit on F^2	1.026	1.017
Final R indices [$I > 2\sigma(I)$]	R_1 = 0.0414 , wR_2 = 0.0836	R_1 = 0.0565, wR_2 = 0.1440
R indices (all data)	R_1 = 0.0553, wR_2 = 0.0927	R_1 = 0.0855, wR_2 = 0.1599

9-coordinated polyhedron between a spherical capped square antiprism (CSAPR-9, CShM = 0.0512) and a tricapped trigonal prism (JTCTPR-9, CShM = 0.0517) as calculated by SHAPE 2.1. Meanwhile, the distorted square pyramidal of Zn^{2+} is completed by 4-nitryl-salicylaldimine and one etheric oxygen of one $\mathbf{L}^{2-}$, one phenoxy oxygen of salicylamide from another $\mathbf{L}^{2-}$ according to the Hard-Soft Acid Base theory (Fig. 1b and Table S3). Consequently, a tetranuclear cluster with two $Zn^{2+}$and two Tb^{3+}bridged by two in-situ generated μ_3- OH is encapsulated between two $\mathbf{L}^{2-}$, affording a sandwich-like tetranuclear Zn^{II}-Tb^{III} compound with metal-ligand ratio of 1 Zn : 1 Tb : 1**L**as depicted in Fig. 1c. Notably, the use of acetonitrile as the sole solvent results in a structure where the metal centers are tightly shielded by two chelating-bridging $\mathbf{L}^{2-}$ and four bidentate NO_3^-. This arrangement leaves the nitro and amide N-H groups of two $\mathbf{L}^{2-}$ oriented outward (Fig. 1d).

Upon introduction of methanol into the reaction solvent system, flake crystals which are quite different from **ZnTb-1** and denoted as **ZnTb-2** are afforded. Structural analysis indicates that **ZnTb-2** crystallizes in the $P2_1/c$ space group. The unit cell of **ZnTb-2** is composed of two molecules with negligible structural difference observed only for bond lengths and angles (Table S2). For each molecule, three crystallographically unique Zn^{2+} and two Tb^{3+} are assembled through salicylamide and 5-nitrosalicylimide groups from four different deprotonated $\mathbf{L}^{2-}$, affording a metal-ligand ratio of 3 Zn : 2 Tb : 4**L**. As highlighted in Fig. 2a, $\mathbf{L}^{2-}$ in **ZnTb-2** is also in open-hoop configuration but with the other two new coordination modes. One adopts a μ_3-**L**-κO1:κO2:κO4:κ N2:κO5 coordination pattern (Mode II) with the 5-nitrosalicylaldimine and neighbouring etheric oxygen chelated by one Zn^{2+}, and the salicylamide chelated by one Tb^{3+} and further anchored by one Zn^{2+}. Notably, 1,2-bis(2-ethoxy)ethyl spacer herein is highly compressed than that in **ZnTb-1** with the average N_{amide}•••N_{imine} distance of 4.512 Å, the average angle between the two benzene rings of 23.998°, and the average centroid distance between the two benzene ring of 5.208 Å. The other $\mathbf{L}^{2-}$ adopts the coordination mode of μ_3-**L**-κ^{2}O1:κO2:κN2:κO5 (Mode III) to chelate one Zn^{2+} by the 5-nitrosalicylaldimine with another Zn^{2+} and one Tb^{3+} clamped by the salicylamide. By contrast, 1,2-bis(2-ethoxy)ethyl spacer in Mode III is extended similarly with that in Mode I, with the average N_{amide}•••N_{imine} distance of 7.179 Å, the angle between the two benzene rings of 44.411°, and the centroids distance between the two benzene rings of 9.252 Å. As shown in Fig. 2b, the 9-coordinated sphere for each Tb^{3+} is fulfilled with two salicylamide group of two $\mathbf{L}^{2-}$ in both mood II and III, two bidentate NO_3^-, leaving the ninth coordination site for a water molecule. The coordination polyhedron of four Tb^{3+} ions are close to the Spherical Tricapped Trigonal Prism (TCTPR-9) with CShM values calculated to be 0.0604, 0.0583, 0.0562 and 0.0533 respectively (Table S3). The bond lengths of Tb–O are in the range of 2.293(4) – 2.541(4) Å. Due to the bridge of four

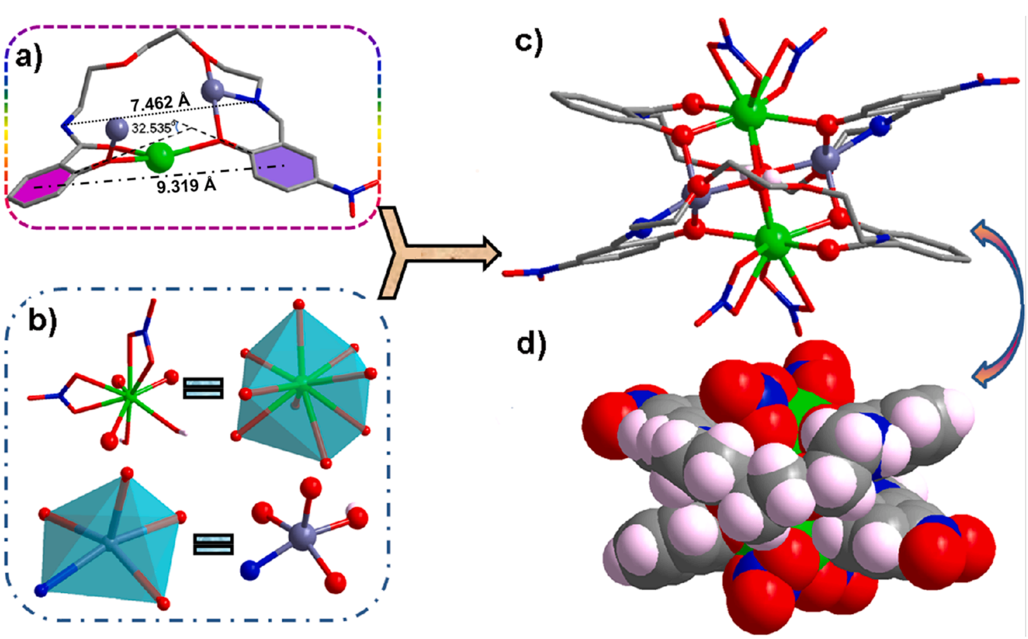


Fig. 1 a). The coordination mode and configuration of $\mathbf{L}^{2-}$ in **ZnTb-1** with hydrogen atoms omitted for clarity; b)The coordination polyhedron of Tb^{3+} and Zn^{2+} in **ZnTb-1**; The molecular structure of **ZnTb-1** with atom model in ball and stick (c) and spacefilling (d). (Symmetry codes:x, -y, -z; Color modes: green, Tb; red, O; blue, N; gray, C; blue gray, Zn).

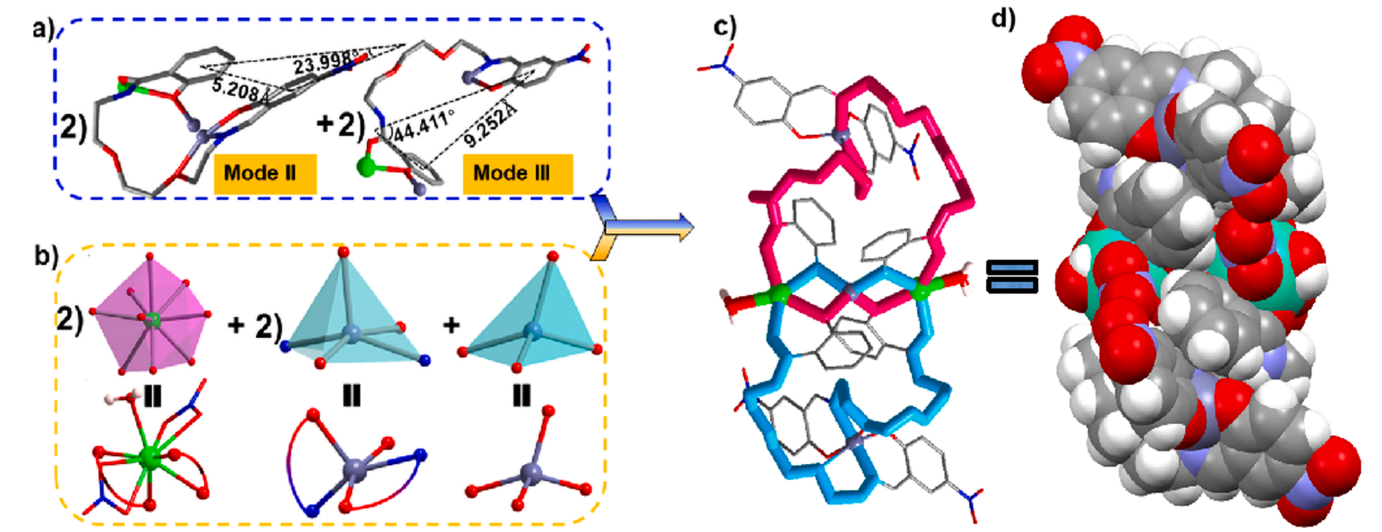


Fig. 2 a). The ligand L^{2-} in coordination mode II and III; b) the coordination sphere and polyhedron of Zn^{2+} and Tb^{3+}; The main architecture of **ZnTb-2** in stick (c) and spacefilling mode (d) (Color modes: gray, C; red, O; blue, N; green, Tb; blue gray, Zn).

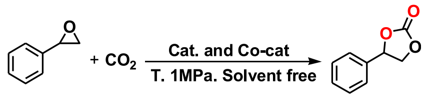
phenoxy oxygen from four salicylamide moieties in four L^{2-} in both Mode II and III, two Tb^{3+} centers are connected by one Zn^{2+} and positioned in the center of the molecule. Meanwhile, the two penta-coordinated square pyramid spheres of Zn^{2+} at both ends are completed by one salicylimine and etheric oxygen of one L^{2-} in Mode II, another salicylaldimine of L^{2-} in mood III. With penta-coordinated Zn^{2+} being node, one L^{2-} in Mode II and one L^{2-} in Mode III are connected to give a large open-loop which is ringclosed to result a 27-atom ring. Furthermore, two such circular rings constituted by two L^{2-} and two Zn^{2+} are interlocked by two Tb^{3+} to afford a much larger Zn^{II}-Tb^{III} heterometallic interlocked cycle as demonstrated in Fig. 2c. Compared with **ZnTb-1**, two bidentate NO_3^- coordinated to the same Tb^{3+} in **ZnTb-2** are opposite rather than adjacent. In addition, the distance of two Tb^{3+} spaced by one Zn^{2+} and two phenoxy oxygen in **ZnTb-2** is much longer than that in **ZnTb-1** where two Tb^{3+} are bridged only by oxygen. Hence, **ZnTb-2** is much loose with more exposed metal centers, amide NH Lewis base sites and two irregular confined cavities (Fig. 2d). As shown in Table S3, the Zn–N bond lengths are in a range from 1.983(5) Å to 2.009 (6) Å and the Zn–O bond lengths range from 1.945(3) to 2.438(5) Å. Compared the three complexes we have obtained based on L^{2-} (**Tb-2 [c]**, **ZnTb-1** and **ZnTb-2**), it is obvious that the incorporation of $Zn^{2+}$and the variation of reaction solvent system play the key role in coordination mode of L^{2-} and sequently the molecular architecture. These observations could be reasonable in that acetonitrile, being a non-protonic solvent, can not stabilize L^{2-} through hydrogen bonding, and the Lewis base site in L^{2-} has a higher degree of exposure to favor metal binding, however, methanol, as a protonic solvent, solvates L^{2-} via H–bonds, thereby reducing the electron-donating ability of its lone pairs. Furthermore, the significant structural divergence from previously reported Zn^{II}-Tb^{III} heterometallic compounds of this kind of ligands reaffirms the decisive influence of the central linker and the terminal functional groups (salicylamide and salicylaldimine) on the final architecture achieved through coordination-driven self-assembly [21,22]

FTIR and TGA Study The structures of **ZnTb-1** and **ZnTb-2** were further elucidated by comparing their IR spectra with that of the free ligand $\mathbf{H_2L}$, which confirmed metal coordination and highlighted key structural differences between them. As shown in Fig. S2, the spectrum of $\mathbf{H_2L}$ displays broad bands at 3449 and 3375 cm^{-1}, attributed to the O–H stretching vibration of the phenolic hydroxy group and the N–H stretches of the secondary amide. The absorptions at 1634 and 1614 cm^{-1} are assigned to the stretching vibrations of the amide carbonyl ($C=O$) and the imine ($C=N$) groups, respectively, while the bands at 1313 and 1132 cm^{-1} correspond to C–O stretching vibrations of the phenolic (Ar–O) and aliphatic ether (C–O–C) groups. The coordination of $\mathbf{H_2L}$ is confirmed by the negative shifts in the amide $C=O$ (from 1634 cm^{-1} to 1612/1614 cm^{-1}) and imine $C=N$ (from 1614 cm^{-1} to 1558 cm^{-1}) stretching vibrations, confirming the involvement of both the carbonyl oxygen and imine nitrogen in coordination. Additional coordination evidence comes from the presence of Zn–O and Tb–O vibrations, observed at 503 and 444 cm^{-1} in **ZnTb-1**, and at 499 and 439 cm^{-1} in **ZnTb**-2. The presence of bidentate nitrates are evidenced by four strong absorption bands at 1480, 1245, 1021, and 821 cm^{-1} for **ZnTb-1** and at 1485, 1250, 1050, and 817 cm^{-1} for **ZnTb-2**. These are attributed to the vibrational modes of the bidentate nitrate ion, specifically the N–O asymmetric stretch (ν_{as}), N–O symmetric stretch (ν_s), N–O stretch (ν) and O–N–O out-of-plane bend, respectively [23]. A key distinction between the two complexes lies in the high-frequency region: in **ZnTb-1**, the broad feature around ~3450 cm^{-1} is consistent with a μ_3-OH-bridged cluster structure, where amide N–H groups are involved in strong N–H···O_{ether} hydrogen bonding. In contrast, **ZnTb-2** shows a well-resolved doublet at 3452 and 3385 cm^{-1}, characteristic of the symmetric and asymmetric O–H stretching vibrations of coordinated and hydrogen-bonded crystalline water molecules [24]. The thermogravimetric experiments were carried out at 30 ~ 800 °C under N_2 atmosphere. As overplayed in Fig. S3, the desolvation of six crystalline acetonitrile in **ZnTb-1** (*Observed*: 13.54 %; *Calculated:* 13.61 %) is completed below 102 °C. With additional thermal decomposition, there is a brief pause and then stability. The structure begins to collapse after 192 °C which was ascribed to the degradation of nitrates and $\mathbf{H_2L}$. For **ZnTb-2**, the initial weight loss below 99 °C (*Observed*: 4.08 %; *Calculated:* 3.91 %) corresponds to the departure of two crystalline acetonitrile and one crystalline water molecule. And rapid decomposition occurs after 185 °C due to the structural breakdown.

3.1. Catalytic performance study

Driven by their heterometallic nature and potential catalytic sites, the catalytic performance of **ZnTb-1** and **ZnTb-2** were evaluated. Initial experiments were conducted at 1 MPa CO_2 pressure using 10 mmol styrene oxide, 0.01 mol % of activated catalyst (16 mg **ZnTb-1** or 25 mg **ZnTb-2**) and 0.75 mol % (24 mg) n-Bu_4NBr as co-catalyst, following conditions of similar systems [18]. Conversion efficiency was determined by ^{1}H NMR (Table 2). **ZnTb-1** yielded 52 % styrene carbonate at 120 °C after 1 h (Entry 1, Fig. S4). In contrast, **ZnTb-2** exhibited superior

Table 2
The catalytic performance for cycloadditionof CO_2 with styrene oxide under different experimental conditions.[a].



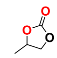
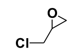
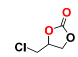
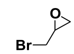
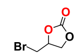
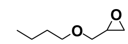
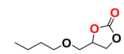
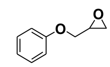
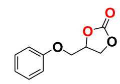
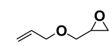
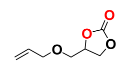
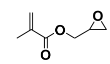
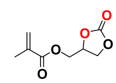
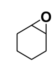
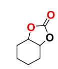
Entry	Cat.	T(°C)	Time(h)	Yields(%)[d]
1	**ZnTb-1**	120	1	54(± 3)
2	**ZnTb-2**	120	1	97(± 2)
3	**ZnTb-2**	90	1	88(± 3)
4	**ZnTb-2**	60	1	52(± 1)
5	**ZnTb-2**	30	1	23(± 4)
6	**H_2L**	120	1	46(± 3)
7	**Tb-2 [*c*]**	120	1	54(± 3)
8	**ZnTb-2**	30	5	49(± 3)
9	**ZnTb-2**	30	10	71(± 2)
10	**ZnTb-2**	30	15	94(±2)
11	**ZnTb-2**	30	20	95(± 2)
12[b]	**ZnTb-2**	30	15	91(± 3)
13[c]	**ZnTb-2**	30	15	89(± 2)

Reaction conditions: styrene oxide (10 m mol), CO_2 1 MPa, 0.75 mol % n-Bu_4NBr, solvent-free. a)Excluding the air inside the reaction system b) Not excluding the air inside the reaction system; c) Addition of 0.2 mL deionized water; d) Yields were determined by ^{1}H NMR.

catalytic performance with a 97 % yield under identical conditions (Entry 2, Fig. S5). This significant difference suggests that the specific structural features, rather than the mere presence of heterometallic sites, are the primary determinant of catalytic activity. Therefore, we selected **ZnTb-2** for further reaction optimization. Lowering the temperature resulted in reduced activity (Entry 3–5: 88 % at 90 °C, 52 % at 60 °C, and 23 % at 30 °C), while elevating the temperature to 150 °C did not significantly enhance the yield. To assess the contribution of the organic ligand and metal centers, control experiments were performed using **H_2L** and its Tb^{3+} compound, **Tb-2 [*c*]**, which features relatively small interlocked cycles at 120 °C. As shown in Entry 6 and 7 of Table 2, **H_2L** and **Tb-2 [*c*]** afforded moderate yield of 46 % and 54 % respectively. The comparable performance of **H_2L,Tb-2 [*c*]** and **ZnTb-1** indicate that the heterometallic Lewis acid sites in **ZnTb-1** contribute little to the reaction. Instead, the Lewis base site (amide N atom) in **H_2L** appears to facilitate the CO_2 cycloaddition, likely due to the presence of multiple supramolecular chemical sites. From a green chemistry perspective, **ZnTb-2** was further evaluated at 30 °C (Entry 5, Entries 8 – 11, Table 2). Prolonging the reaction time from 1 to 20 h progressively increased the yield (23 % → 49 % → 71 % → 94 % → 96 %). A 94 % yield was achieved within 15 h with only marginal improvement upon further extension. Thus, the optimal reaction conditions could be 10 mmol styrene oxide, 0.01 mol % **ZnTb-2**, 0.75 mol % n-Bu_4NBr, at 30 °C for 15 h and solvent free. Notably, the conversion was largely unaffected by air or water (Entries 12 – 13, Table 2), suggesting its strong potential for CO_2 capture from flue gas.

To evaluate the universality of this catalyst, a range of epoxy compounds bearing different substituents were conducted under our established reaction protocols. As illustrated in Table 3, **ZnTb-2** exhibited excellent catalytic efficiency toward the CO_2 cycloaddition with various epoxides. The reaction with propylene oxide achieved near-quantitative yield of the corresponding cyclic carbonate within 15 h (entry 1, Fig. S6). The high yield is likely attributable to its small molecular size which facilitates access to the catalytic sites. To probe the influence of electronic effects, epichlorohydrin, epibromohydrin, butyl glycidyl ether, glycidyl phenyl ether, allyl glycidyl ether and glycidyl methacrylate were tested (Entries 2~7, Fig. S7-12). These substrates yielded their respective cyclic carbonate in 94 % – 97 % conversion with high than 99 % selectivity, indicating that the electronic properties of the substituents have minimal effect on the reaction. Finally, cyclohexene oxide was employed to explore steric effect, and a moderate yield of 62

Table 3
The catalytic performance of **ZnTb-2** for cycloadditionof CO_2 with various epoxy compounds.

Entry	Substrate	Product	[a]Yield (%)	[b]Selectivity	[c]TON
1			99(± 4)	100	9897
2			97(± 3)	99	9668
3			94 (± 3)	99	9412
4			96(± 3)	99	9588
5			95 (± 3)	99	9478
6			96 (± 3)	99	9578
7			96 (± 3)	99	9589
8			62 (± 2)	99	6186

[a]Reaction conditions: 10 mmol epoxide, 0.01 mol % catalyst, 0.75 mol % n-Bu_4NBr as the co-catalyst, 30 °C, 15 h, 1 MPa. [b] Yields are determined by ^{1}H NMR. [c]TON = (moles of product)/(moles of catalyst).

% is afforded (Entry 8, Fig. S13). This lower reactivity can be ascribed to the significant steric crowding around the epoxide rings of cyclohexene oxide.

Recyclability is a crucial attribute for evaluating the practical application of a catalyst. The tested catalyst, **ZnTb-2**, could be easily separated from the reaction mixture by adding CH_2Cl_2 to precipitate and followed by centrifugation. This process yielded a recovery of approximately 22 mg with the loss of 9 %. The PXRD pattern of the regenerated **ZnTb-2** sample closely matched that of the as-synthesised (Fig. S14), confirming that the catalyst's structural integrity was preserved upon regeneration. This was further supported by the identical emissive behavior of the regenerated sample and the pristine samples (Fig. S15) and infraed spectroscopy, which showed negligible differences between them (Fig. S16). Furthermore, the recovered **ZnTb-2** maintains high catalytic efficiency over four reaction cycles without an appreciable loss in activity (Fig. S17), indicating excellent recyclability.

The catalytic efficiency of **ZnTb-2** for the CO_2-to-epoxide cycloaddition is better than or comparable to several reported Zn^{II}-Ln^{III} heterometallic catalysts when considering loading amounts, CO_2 pressure, temperature and reaction time. For example, the helicate $Zn_4Tb_3L_4$ constructed from a helical ligand H_3L with 2-hydroxypropyl terminated by salicylamide and 3-methoxysalicylaldimine achieves 96 % conversion, but require a higher catalyst and co-catalyst loading [18]. Similarly, the heterometallic cage $Zn_3Nd_4L_6$ based on a similar ligand with ethyl terminated by salicylamide and 4-ethylsalicylaldimine shows good activity for propylene oxide (95 %) but lower conversions for styrene oxide (66 %) under reaction condition of 10 mmol epoxide, 0.01 mol % catalyst, 0.8 mol % n-Bu_4NBr, 1 MPa CO_2, 80 °C and 6 h [19]. Two unique gridlike 3d-4f heterometallic structures, $Ln_8Zn_4L_6$ (Ln = Eu(1), Tb(2), H_4L = 2,2′-oxybis (N'-((E)-2-hydroxy-3-methoxybenzylidene) acetohydrazide)), showed a 93 % percentage yield at 0.01 mol % catalyst, 0.5 mmol % co-catalyst TBAB, 1 MPa CO2, 120 °C and 1 h [25]. All these discrete Zn^{II}-Ln^{III} species underperform compared to **ZnTb-2** under identical conditions (Entry 2, Table 1, Entry 1, Table 2). Recently reported Zn^{II}-Ln^{III} heterometallic coordination polymers,

$[Zn^{II}Ln^{III}(bipydc)_2(HCOO)(H_2O)_3]{\bullet}2H_2O$, Ln = Pr(I), Nd (II), Sm(III) and Eu(IV), afford inferior catalytic efficiency for allyl glycidyl ether with a percentage yields of over 80 %, percentage conversion of 85 – 90 %, turn-over number of 2088 – 2260 at 1.6 mol % n-Bu_4NBr, 0.040 mol % catalyst, 90 °C, 4 h and ambient pressure [26]. By contrast, Zn^{II}-Ln^{III} heterometallic MOFs exhibit high activity but require higher catalyst/co-catalyst loadings or longer reaction times. NUC-62a, obtained 97 % conversions of styrene oxide with 0.5 mol % of catalyst, 5 mol % n-Bu_4NBr at 75 °C and 6 h [27]. And a Zn^{II}-Tb^{III} MOF, NUC-7, shows 37 ∼ 99 % conversions at 20 mmol epoxides, 5 mol % n-Bu_4NBr, 0.2 mol % NUC-7 catalyst, 1 atm CO_2, 80 °C and 24 h [28]. Therefore, **ZnTb-2** could be a better catalyst, as it operates effectively under milder conditions with lower catalyst and co-catalyst requirements and the ease of preparation and regeneration.

3.2. Catalytic mechanism study

As recently demonstrated, coordination environment engineering is a key strategy for optimizing the electronic configurations of metal-based catalytic centers, enhancing active site density, and improving electron transfer kinetics [29]. Given the X-ray elucidated structure of **ZnTb-2** and related literatures [30] the excellent catalytic property of **ZnTb-2** could be rationalized by the three key factors: 1) the labile Tb-O_{water} bonds render Tb^{3+} potential Lewis acid sites for substrate activation, while the no-coordinated amide-NH acted as potential Lewis base sites to capture and activate CO_2; 2) The confined cavities prolong the contact time between the substrate and catalyst; 3) A synergistic effect, originating from the distinct electronic properties of Zn^{2+} and Tb^{3+} enhances the catalytic activity. To gain clearer insights, DFT calculations were first conducted to determine the optimized geometries of **ZnTb-1, ZnTb-2, Tb-[2]*c*** and their PO complex analogue (**ZnTb-1**@PO, **ZnTb-2**@PO and **Tb-[2]*c***@PO) (see the Supporting Information for details). As depicted in Figs. S18, S19 and Table 4, the largest positive Mulliken charge resides on Tb element in all the three structures. And in the case of amide hydrogen, those for **ZnTb-2** show largest positive charge. As depicted in Fig. S20, **ZnTb-2** exhibits a preferential binding of PO to its Tb^{3+} centers with short PO···Tb^{3+}bond distances of 2.5273 Å and 2.5423 Å. In contrast, **Tb-[2]*c***@PO and **ZnTb-1**@PO show much weaker interactions with significantly longer distances ranging from 5.3372 to 6.1229 Å. This suggest a plausible mechanism: The confined cavities of **ZnTb-2** first enrich CO_2 and epoxide substrates. The coordination bonds between H_2O and Tb^{3+} then dynamically dissociate, exposing Tb^{3+} Lewis acid to bind and activate the epoxide. Meanwhile, CO_2 is activated by amide NH via hydrogen bonds. The nucleophilic Br^- from the co-catalyst TBAB then attacks the activated epoxide to form a key Tb^{3+}-coordinated bromoalkoxide intermediate. The nucleophilic alkoxide is rapidly attacked by CO_2, forming an acyclic carbonate. Finally, the cyclization step releases Br^- and forms cyclic carbonate product, regenerating **ZnTb-2** for the next catalytic cycle.

To verify this, DFT calculation are performed with propylene oxide (PO) as the epoxide model, Br^- instead of TBAB cocatalyst and half of **ZnTb-2** molecule to reduce the calculation burden [31]. The optimized structures of the intermediates and transition-state, along with the corresponding potential energy profiles, are shown in Scheme 2a and b. The sum of the relative energies of **ZnTb-2** and substrates (CO_2 + PO) are set as zero-energy reference. As illustrated, the reaction is initiated by thermodynamically favorable exchange of coordinated water molecules with PO, which result in a slight free-energy decrease of 0.7 kcal·mol^{-1}, forming the Tb^{3+}-coordinated PO intermediate (**Int-0**) with a O-Tb bond distance of 2.55 Å. This coordination polarizes C-O bonds of PO molecule, making it more susceptible to nucleophilic attack. In the subsequent step, Br^-, from TBAB, attacks the less-hindered methylene carbonatom (β-C) of the coordinated PO in **Int-0**. This barrier-free step is highly exergonic, with a significant energy release of 45.6 kcal·mol^{-1}, leading to the low energy intermediate **Int-1**, indicating that PO activation occurs readily. The nucleophilic attack by Br^- opens the three-membered epoxide ring via transition state **TS-1**, forming intermediate **Int-2**. This step involves C-Br bond formation (2.53 Å) and cleavage of C-O bond (lengthened from 1.43 Å to 1.86 Å) and requires a substantial activation barrier of 20 kcal·mol^{-1}. Upon ring opening, the Tb-O_{PO} shortens to 2.24 Å and the C-Br bond shortens to 2.06 Å, accompanied by an energy release of 6.7 kcal·mol^{-1}. Intermediate **Int-3** is then formed through interaction between electrophilic CO_2 and electron-rich ether oxygen of the ring-opened PO, with a relative low energy of -31.1 kcal·mol^{-1}. The insertion of CO_2 into the Tb-O_{PO} bond through electrostatic interaction yields intermediate **Int-4** (-35.6 kcal·mol^{-1}), which proceeds via transition state **TS-2** (-26.5 kcal·mol^{-1}). Ring closure is then facilitated through transition state **TS-3**, which requires an energy barrier of 7.9 kcal·mol^{-1}. During this step, Tb-O_{PO} bond lengthens from 2.40 Å to 2.51 Å. The nucleophilic oxygen from the inserted CO_2 displaces the Br^- anion, causing the C_{PO}-Br bond to lenghten from 2.37 Å to 3.43 Å, and forming intermediate **Int-5.** This is also barrier-free with a free-energy decrease of 26.1 kcal·mol^{-1}. Finally, the dissociation of cyclic carbonates regenerates **ZnTb-2** catalyst, completing the catalytic cycle. As discussed, the high energy difference between **Int-2** and **TS-1** is associated with the breakage of β-C-O bonds. This align with the reported studies, identifying the epoxide ring-opening as the rate-determining step with a high energy barrier in cycloaddition reactions [32,33].

4. Conclusion

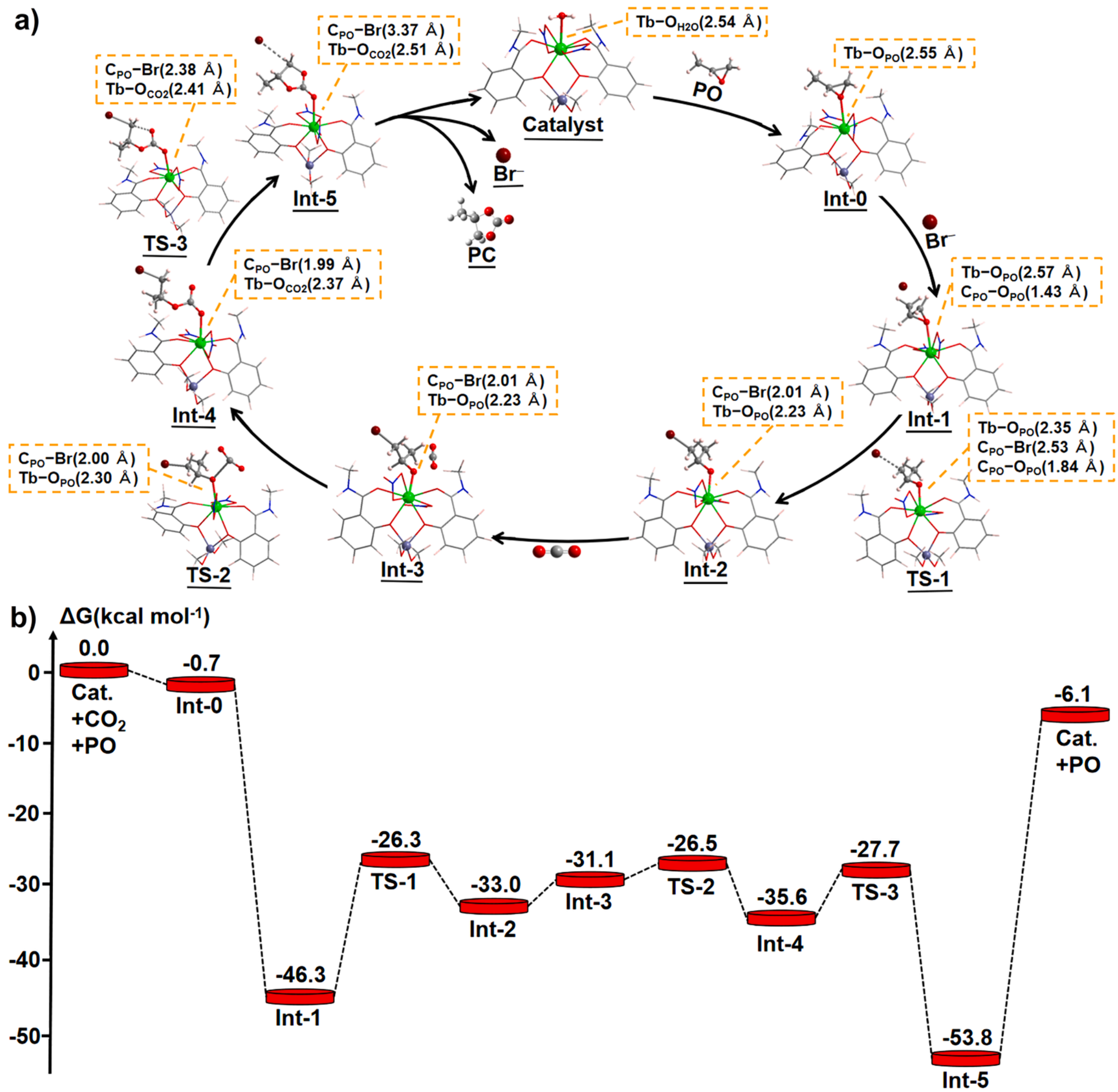
In conclusion, this work successfully demonstrates that precise solvent-directed synthesis can be used to engineer novel Zn^{II}-Tb^{III} heterometallic complexes with distinct structures, which in turn govern their catalytic performance. The interlocking cyclic compound, **ZnTb-2**, proved to be a highly efficient catalyst for CO_2 cycloaddition under mild conditions, outperforming the sandwich-structured **ZnTb-1**. This is attributed to **ZnTb-2**'s unique structural features, a confined cavity, more exposed metal centers and accessible Lewis base sites, which collectively provide an optimal environment for substrate interaction. Therefore, This study underscores the importance of precise heterometallic complex engineering in optimizing CO_2 conversion, providing valuable insights for the design of highly efficient, sustainable catalytic systems.

ASSOCIATED CONTENT

A brief statement of materials and physical measurements, single crystal X-ray diffraction study, important bond lengths, shape calculation results, IR spectra, PXRD patterns, TGA curves, emission spectra, ^{1}H

Table 4
The Mulliken charges values on the most important atoms in **ZnTb-1, ZnTb-2 , Tb-2[*c*] , ZnTb-1**@PO, **ZnTb-2**@PO and **Tb-2[*c*]** @PO. .

AtomMulliken charge	Tb1	Tb2	Zn1	Zn2	Zn3	NH1	NH2	NH3	NH4
ZnTb-1	1.303	1.301	1.134	1.136	–	0.301	0.301	–	–
ZnTb-2	1.362	1.343	1.165	0.840	0.853	0.363	0.359	0.310	0.305
Tb-2[*c*]	1.611	1.609	–	–	–	0.253	0.251	–	–
ZnTb-1@PO	1.279	1.325	1.138	1.140	–	0.302	0.301	–	–
ZnTb-2@PO	1.451	1.420	1.182	0.840	0.852	0.365	0.357	1.279	1.279
Tb-2[*c*] @PO	1.655	1.631	–	–	–	0.252	0.251		



Scheme 2. a) Mechanistic pathways showing the optimized structures of proposed intermediates and transition states involved in the cycloaddition of CO_2 with PO catalyzed by **ZnTb-2**; b) Energy profile piagram of the intermediates(**Int**) and transition states (**TS**) in **ZnTb-2** catalyzed CO_2 cycloaddition reaction calculated by DFT.

NMR spectra of H_2L and that for the CO_2 cycloaddition reaction mixture. Crystallographic files in CIF format have been deposited with the Cambridge Crystallographic Data Center with deposition numbers CCDC 2484801, 2171,977.

CRediT authorship contribution statement

Xue-Qin Song: Writing – review & editing, Supervision, Investigation, Funding acquisition, Data curation, Conceptualization. **Peng-Li Zhao:** Writing – original draft, Methodology, Investigation. **Yi-Yi Yang:** Visualization, Validation, Data curation. **Peng-Fei Qiu:** Validation, Investigation, Data curation.

Declaration of competing interest

The authors declare no competing financial interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.molstruc.2025.144860.

Data availability

The data that has been used is confidential.

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